MODEL X 2.0

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**PHASE 1 · PROJECT PROPOSAL**

**Team name** Silent Node

IEEE Computational Intelligence Society Student Chapter, IIT

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|---|---|---|
| ☐ | **01** | Cost of living and everyday economics |
| ☐ | **02** | Climate, environment and disaster risk |
| ✓ | **03** | **Health and wellbeing** |

## 1. Problem statement and definition

Dengue represents Sri Lanka's most consequential recurring public-health emergency, consistently outrunning the response mounted against it. The country's largest recorded outbreak, in 2017, produced 186,101 reported cases and 440 deaths (Tissera et al., 2020), peaking at over 10,000 cases in a single week in mid-July. Transmission has remained endemic at scale since: verified district-level datasets demonstrate roughly 49,874 cases in 2025 and 30,485 cases in the first 20 weeks of 2026, with Colombo, Gampaha, Kalutara, Galle, Matara, and Ratnapura persistently dominant.

The affected population comprises residents across all 26 dengue-reporting units (the 25 administrative districts plus the separately reported Kalmunai health zone)—representing approximately 21.78 million individuals exposed to *Aedes aegypti* and *Aedes albopictus* in urban and peri-urban environments. The impact is measured in mortality, elevated hospital occupancy during the June–July and December–January monsoon peaks, economic losses, and substantial burdens placed on approximately 350 Medical Officer of Health (MOH) divisions.

Currently, district vector-control teams allocate larval-source-reduction and fogging resources based on preceding weekly case reports—a reactive indicator. Because precipitation and temperature influence vector abundance 5 to 12 weeks prior to reported human transmission, reactive deployments occur after transmission has already escalated. Given that operational interventions require 2 to 4 weeks of lead time for effective deployment (Hussain-Alkhateeb et al., 2021), surveillance signals utilized today arrive too late to inform proactive strategy.

## 2. The user

The primary target user is a district vector-control officer or Medical Officer of Health (MOH) in Sri Lanka—an operational administrative role rather than the general public. This stakeholder oversees resource management, including dedicated inspection and fogging crews, larvicide allocation, and weekly operational planning.

Operational Decision: Allocate field teams across MOH divisions for a 2-to-4-week forward window, determining priority zones for inspection, source reduction, pre-positioning of larvicide, and hospital liaison.

Frequency: Weekly, aligned with Sri Lanka's national surveillance reporting cycle.

Current Baseline Information: Preceding weekly district case counts published by the Epidemiology Unit alongside local administrative context. This serves as a lagging, aggregated metric that shifts upward only after local transmission is established.

Proposed System Impact: Decision-makers start each planning cycle with a ranked, probabilistic trajectory for each district over a 4-week horizon—identifying low-current-incidence areas with high statistical likelihoods of exceeding alert thresholds. This enables proactive resource deployment

ahead of epidemic curves. The system functions explicitly as a planning decision-support framework, not a clinical diagnostic tool.

## 3. Literature review

Existing Sri Lanka-specific research establishes predictive signal viability but lacks operational planning integration:

- Withanage et al. (2018, Parasites & Vectors) developed SARIMA, Poisson, and negative-binomial models for Gampaha district, demonstrating that lagged precipitation (4–10 weeks) and temperature (4–8 weeks) improve prediction accuracy. However, this study focused on a single district using pre-2020 static data without operational deployment.
- Liyanage et al. (2016, IJERPH; 2022, Lancet Planetary Health) established ENSO and local meteorological effects on dengue dynamics in Kalutara district and island-wide, identifying 3-to-6-month teleconnections. These studies remain explanatory rather than operational.
- Tissera et al. (2020, Emerg Infect Dis) documented the 2017 epidemic, confirming that outbreaks follow predictable environmental and immunological interactions, and highlighting the delay in traditional response mechanisms.

International research validates the proposed architectural best practices:

- Lowe et al. (2014, Lancet ID; 2016, eLife) demonstrated probabilistic climate-driven dengue risk stratification for Brazil at 1-to-3-month lead times evaluated against seasonal baselines, providing a conceptual framework for this project.
- Johansson et al. (2019, PNAS) evaluated open dengue forecasting challenges (Iquitos, San Juan), establishing that many advanced models failed to consistently outperform seasonal-naive baselines—underscoring the necessity of rigorous benchmarking.
- Bowman et al. (2016) and Hussain-Alkhateeb et al. (2021) validated WHO-TDR's EWARS early warning framework across endemic regions, confirming the 4-to-12-week forecast window as the primary operational target.

Open Data Infrastructures: Clarke et al. (2024) released OpenDengue, while the denguedatahub R package (Talagala) compiles Sri Lanka's district-level weekly series.

Identified Research Gap: While Sri Lanka possesses extensive multi-decade district surveillance data and proven climate predictability, there is no open, reproducible, weekly updated district-level forecasting framework providing actionable risk thresholds directly to operational health officers.

## 4. Proposed solution

We propose an automated, weekly updated, 4-week-ahead dengue risk forecasting framework covering Sri Lanka's 26 health districts. Model outputs are formatted into decision-ready risk tiers—Low, Moderate, High, and Extreme—accompanied by complete predictive confidence intervals and optimized inspection team allocation plans.

Core Intelligence Generated: For each district, the platform provides a probabilistic 4-week trajectory, including explicit threshold-exceedance probabilities (e.g., 'Colombo exhibits a 78% probability of exceeding its historical 95th-percentile weekly threshold within 3 weeks') and early anomaly flags.

Analytical Complexity: The model integrates 19+ years of weekly epidemiological history with lagged climate variables (precipitation lagged 5–12 weeks; temperature and relative humidity lagged 8–14

weeks), models spatial transmission spillover across contiguous districts, overlays static infrastructure vulnerability indices (e.g., municipal piped water coverage), fits overdispersed count distributions per district, and propagates predictive uncertainty.
Fail-Safe Design: Risk tiers include full prediction intervals. Districts with missing or stale inputs are explicitly categorized as 'Insufficient Data' rather than defaulted to low risk. System interfaces maintain prominent non-clinical disclaimers emphasizing decision-support scope.

# 5. Data

All primary datasets have been acquired, parsed, and validated. Empirical figures in this proposal are derived directly from primary data files.

## 5.1 Source and access

Primary Epidemiological Data: The denguedatahub R package (Talagala; CRAN/GitHub), specifically the srilanka_weekly_data structure, provides 26,311 records of weekly district case counts across 26 reporting units from 2006-W52 through 2026-W20 (1,352 district-weeks per annum), compiling official Sri Lanka Epidemiology Unit surveillance reports.

Contextual Data: OpenDengue v1.3 (Clarke et al., 2024) provides national series for Sri Lanka (1965–2024) containing 579 weekly/monthly national records utilized for macro-trend validation.

Meteorological Data: Open-Meteo API (ERA5 reanalysis archive and 14-day weather forecasts) queried at district centroid coordinates. CHIRPS daily 0.05° precipitation data provides high-resolution rainfall fallbacks.

Climate Drivers: NOAA Oceanic Niño Index (ONI) and Bureau of Meteorology Dipole Mode Index (DMI) joined at 3-to-6-month lags to capture macro-climatic teleconnections (Liyanage et al., 2016, 2022).

Spatial Boundaries: OCHA COD-AB Sri Lanka administrative boundaries (HDX) for GIS mapping across 25 administrative districts.

Demographics: OCHA/WorldPop COD-PS 2023 subnational population dataset (lka_admpop_adm2_2023.csv) establishing baseline denominators for per-100,000 incidence calculations.

Infrastructure Data: Department of Census and Statistics CPH-2024 Table A5 ('Households by main source of drinking water'), quantifying district-level piped water access (38.9% national average vs. 84.4% in Colombo).

Data Governance & Privacy: The pipeline utilizes fully anonymized administrative counts exclusively, ensuring complete privacy compliance and non-clinical boundary adherence.

Data Access Verification (Executed October 17, 2026):

- Successfully parsed srilanka_weekly_data.rda (26,311 records across 26 districts from 2006-W52 through 2026-W20). Figure 1 displays the ingested data.
- Validated OpenDengue v1.3 spatial extracts (~55 MB CSV) to confirm subnational data coverage limits.
- Queried Open-Meteo historical APIs for Colombo coordinates (6.9271°N, 79.8612°E), extracting 692 weeks of meteorological data (2010 to May 2026). Tested live 14-day forward forecast queries.

- Verified active status of the WHO SEARO Dengue Dashboard. Identified bandwidth restriction errors on the primary Ministry of Health portal, establishing the necessity of automated fallback pipelines.

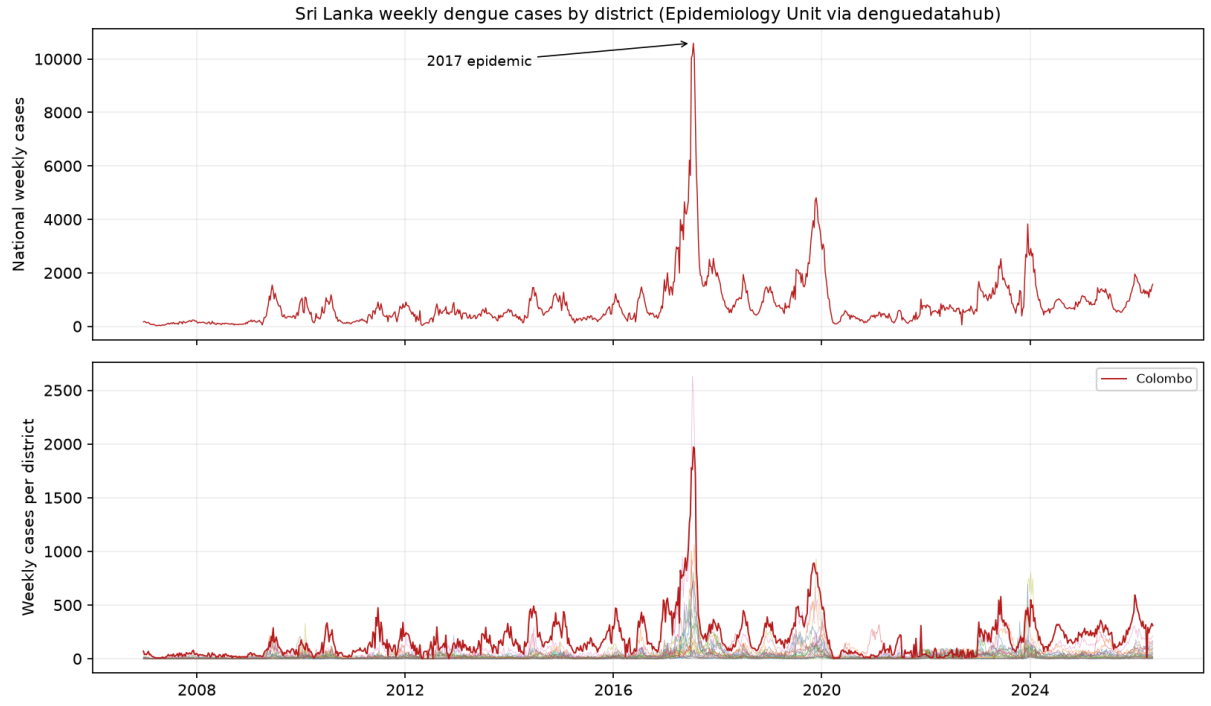



Figure 1. Test-query evidence, plotted directly from the downloaded denguedatahub file (17 Oct 2026): national weekly totals and per-district series, 2006-W52 to 2026-W20. The July 2017 peak (10,590 cases/week) is visible in red.

- Parsed COD-PS 2023 demographic files and CPH-2024 water access tables, successfully extracting district-level covariate vectors.

Data Volume Summary: 20 years × 26 districts × 52 weekly records; 16 years of daily meteorological parameters across 26 district centroids. Historical outbreak years (2017, 2019, 2023) fall entirely within the training set.

## 5.2 Extraction method

Case Data Processing: Automated ingestion of GitHub `.rda` repositories converting records into standardized tabular structures. Live surveillance reports published via weekly Ministry PDFs are parsed using `pdfplumber`, with automated fallbacks to the WHO SEARO dashboard when primary endpoints exhibit service degradation.

Meteorological Pipelines: Daily parameters retrieved via Open-Meteo REST endpoints across 26 district centroids, aggregated to epidemiological weeks, and concatenated with 14-day weather forecasts.

Static Covariate Processing: CPH-2024 tables parsed via `openpyxl` to extract municipal water access proportions per district. Queen contiguity matrices derived from OCHA shapefiles model spatial spillover effects.

Data Cleaning Procedures: Standardized epidemiological-to-ISO week alignment; district nomenclature normalization; missingness handling; lag-aware training windowing (excluding partial

data under 2 weeks old); and administrative boundary reconciliations between municipal and regional health reports.

### 5.3 Temporality

The framework combines 20 years of historical surveillance data with automated weekly ingestion and 14-day meteorological forecasts. This architecture addresses operational requirements: static historical models lack current sensitivity, while purely real-time feeds lack historical context. Combining these streams supports 2-to-4-week advance planning.

Refresh Cadence: Weekly execution aligned with Epidemiology Unit releases. If primary data sources drop offline, the system falls back to preceding valid periods, expands predictive variance bounds, and displays a prominent data-staleness banner.

Static Parameter Handling: Demographics and water infrastructure data are updated on multi-year census cycles and modeled as fixed baseline covariates, reflecting conditions from the 2024 census.

## 6. Approach and methodology

Primary Model: Negative-binomial Generalized Additive Models (GAMs) with distributed lag structures per district, combined with a LightGBM ensemble module. Model features include autoregressive terms (1–6 weeks), harmonic seasonal components, lagged meteorological metrics (precipitation lagged 4–12 weeks; temperature/humidity lagged 8–14 weeks), and monsoon phase flags. The negative-binomial formulation addresses count data overdispersion to output full probability distributions.

Climate Driver Integration: Monthly ENSO (ONI) and IOD (DMI) indices lagged 3 to 6 months incorporate macro-climate signals into the feature matrix, subject to ablation testing against baseline models.

Empirical Feasibility: Cross-correlation analysis of 692 weeks (2010–2026) of Colombo case data against Open-Meteo weather parameters confirms peak precipitation correlation at 5–6 week lags ($r \approx 0.19$) and temperature at 10–14 week lags ($r \approx 0.13$), with peak seasonal transmission aligned with the southwest monsoon (July mean ~410 cases/week vs. March mean ~145 cases/week).

Alternative Approaches Evaluated: Mechanistic SIR models were omitted due to unidentifiable vector-density and serotype parameters at district scales; deep learning architectures (LSTM/Transformers) were excluded due to overfitting risks on short, noisy time series; search-query nowcasting was rejected due to unstable localized search volume.

Spatial spillover: transmission crosses district borders with commuting and short-range mosquito dispersal, so we add neighbour-weighted case counts (row-standardised queen-contiguity weights from the district map) at 1-4 week lags as features. A rising epidemic next door raises a district's forecast — an effect an officer reading only their own district's counts would miss; Liyanage et al. (2016) showed dengue transmission in Sri Lanka is spatially structured at exactly this scale.
Population exposure: district population enters the model as an exposure offset and all tiers are expressed as incidence per 100,000, so large districts do not dominate simply for being large.
Infrastructure: piped-water access enters as a covariate — stored water is a primary Aedes breeding site, and coverage varies sharply across districts (84% of Colombo households use NWSDB piped water versus 39% nationally).

Spatial Spillover & Population Standardization: Neighbor-weighted case counts based on spatial contiguity matrices capture inter-district transmission dynamics. Population offsets convert raw predictions to standardized incidence rates per 100,000 population. Piped-water access proportions incorporate baseline infrastructure vulnerabilities.

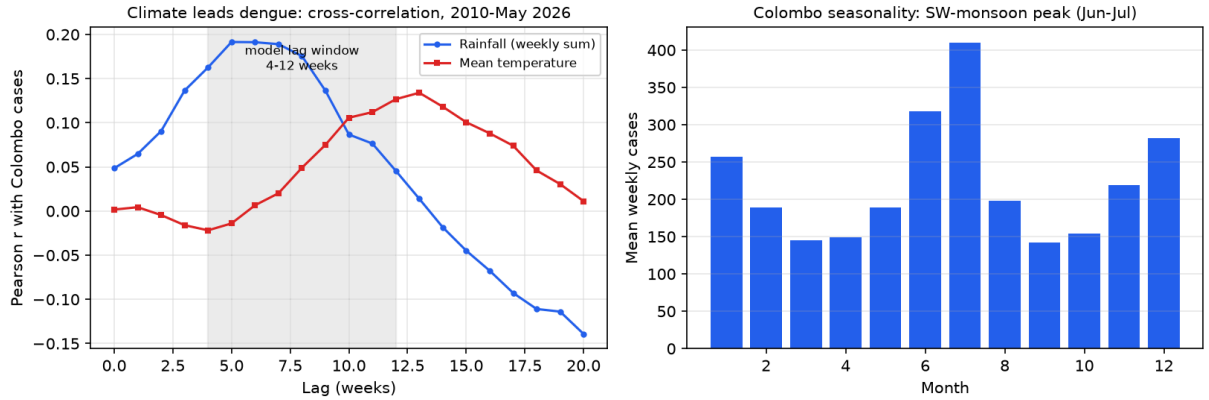



Figure 2. Our own feasibility check (17 Oct 2026): Colombo weekly cases vs Open-Meteo climate, 692 weeks (2010-May 2026). Left: rainfall leads cases by 5-6 weeks, temperature by 10-14 weeks — the model's lag window. Right: the monsoon seasonality the seasonal terms capture.

# 7. System design

## 7.1 Architecture

The framework operates across four decoupled processing stages orchestrated via GitHub Actions batch workflows:

- Data Ingestion: Ingest weekly case reports and meteorological records/forecasts across 26 district centroids; execute schema validation and anomaly checks.
- Feature Store: Align epidemiological timelines, normalize district identifiers, construct autoregressive and lagged climate feature matrices, and compute spatial spillover matrices.
- Forecast Engine: Update district-level models, generate 1-to-4-week predictive distributions, evaluate threshold exceedance probabilities, and assign risk tiers.
- Delivery Layer: Serve findings through an interactive web dashboard and exportable CSV allocation matrices.

## 7.2 Technical stack

Stack Components: Python (`pandas`, `statsmodels`, `scikit-learn`, `LightGBM`) for modeling pipelines; R (`denguedatahub`) for data extraction; DuckDB for analytical storage; GeoPandas for GIS processing; Streamlit for dashboard rendering; and GitHub Actions for automated weekly execution.

## 7.3 Interface and interaction

Dashboard Layout: Choropleth district map color-coded by 4-week risk tier, supported by a structured table displaying predicted case counts, incidence rates per 100,000, 50%/95% confidence intervals, and threshold exceedance probabilities. Includes one-click CSV export for resource planning alongside explicit data-freshness indicators and non-clinical planning disclaimers.

# 8. Implementation plan

Development Roadmap (October 19 – November 1):

- Oct 19–22 — Data Pipeline & Baselines: Finalize automated data extraction, feature engineering pipelines, seasonal-naive baseline models, and rolling backtest frameworks.
- Oct 23–27 — Climate Integration & Model Fitting: Integrate meteorological feature vectors, fit negative-binomial GAMs, execute backtesting against historical outbreak years, and calibrate risk tiers.
- Oct 28–31 — Dashboard & Automation: Construct interactive web interfaces (map, summary tables, CSV exports), integrate data-staleness banners, and configure GitHub Actions automation workflows.
- Nov 01 — Verification & Documentation: Complete technical documentation, performance evaluation reports, executive summaries, and system demonstration media.

Success Criteria: Complete execution of automated weekly pipelines generating calibrated 4-week risk classifications across all 26 districts, demonstrating statistical improvement over seasonal-naive baselines during historical validation periods. Contingency adjustments prioritize primary GAM pipeline delivery over optional optimization modules.

## 9. Evaluation plan

Performance Metric: System accuracy is defined by operational lead-time utility—specifically, outperforming lagging indicators and seasonal baselines in forecasting risk escalation 2 to 4 weeks in advance.

Validation Methodology: Rolling-origin backtesting (training through week t to predict t+1..t+4) across 2022–2026 test windows. Forecast quality is evaluated using Weighted Interval Scores (WIS) for probabilistic distributions (Bracher et al., 2021) and ROC-AUC for high-risk threshold exceedance alerts, benchmarked against seasonal-naive, last-value, and autoregressive baselines.

Fallback Criteria: If climate-driven models fail to demonstrate superior WIS skill over seasonal-naive benchmarks at 2-week horizons, the system defaults to deploying calibrated seasonal-autoregressive models as the operational baseline.

## 10. Risks and contingencies

- Reporting Delays & Partial Data: Recent weekly counts are frequently incomplete. Mitigation: Restrict model training and evaluation to data ≥2 weeks old; model recent periods as partial inputs; and quantify delay sensitivity explicitly.
- Primary Source Instability: Public endpoints may experience bandwidth issues. Mitigation: Maintain dual-path ingestion using official Ministry reports primary feeds and WHO SEARO endpoints as secondary fallbacks, accompanied by clear interface data-freshness stamps.
- Archive Coverage Limits: Historical datasets end at fixed dates. Mitigation: Use denguedatahub records for 2006–2026 model training and integrate active weekly surveillance feeds for real-time operation.
- Meteorological Noise: Reanalysis precipitation estimates may exhibit variance in tropical regions. Mitigation: Use CHIRPS 0.05° daily data as a alternative source and aggregate features over multi-week windows to reduce daily noise.
- Unexpected Epidemiological Shifts: Outbreak dynamics driven by novel viral strain introductions. Mitigation: Output full predictive variance bounds rather than point projections and present threshold exceedance probabilities clearly.

- Resource Constraints: Potential team capacity fluctuations. Mitigation: Implement modular pipeline engineering allowing independent development of ingestion, modeling, and presentation layers.
- Static Covariate Limitations: Demographics and infrastructure parameters reflect census snapshots. Mitigation: Model static covariates as baseline geographic factors and run sensitivity analyses excluding them to evaluate feature importance.

Methodological Risk Statement: The principal analytical risk is that lagged meteorological parameters may yield marginal predictive gain over strong seasonal baselines during non-epidemic periods. Rigorous out-of-sample evaluation will determine feature inclusion.

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## 12. Team details

Roster composition is locked per competition registration specifications.

Team member card — add photo, then: Name: [FILL IN] · Email: [FILL IN] · University: [FILL IN]

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## AI-assistance disclosure (competition rule 06)

Artificial intelligence tools were utilized during proposal development for literature synthesis, language editing, and automated dataset verification scripts (including test queries, downloads, and visual generation). All dataset assessments, analytical decisions, figures, and model formulations were reviewed, verified, and are fully owned by the project team.